WB Report

Sadam Ramakrishna EE23B067

March 4, 2026

1 2.1 Dataset Visualization

I logged five representative samples from each of the ten classes.

In MNIST, certain digits such as 4 and 9 or 3 and 5 appear visually similar due to overlapping stroke patterns. This similarity increases the risk of misclassification because the model must distinguish subtle pixel differences.

In Fashion-MNIST, categories such as Shirt and T-shirt or Sneaker and Ankle Boot exhibit structural similarity. Unlike digits, Fashion-MNIST requires capturing texture and shape variations, making classification more complex.

2 2.2 Hyperparameter Impact Analysis

From the Parallel Coordinates visualization of 130 runs, the learning rate had the strongest effect on validation accuracy. Very high learning rates (0.05) caused unstable training, while moderate values (0.001–0.01) performed best.

Xavier initialization consistently outperformed random initialization. RMSProp and Momentum optimizers generally achieved higher validation accuracy than vanilla SGD. Batch size and number of layers showed comparatively weaker impact.

Best run configuration:

```
"hidden_size": [256,128,64], "activation": "sigmoid", "loss": "cross_entropy"
"weight_init":"xavier", "batch_size":32, "learning_rate":0.005, "optimizer":"rmsprop"
```

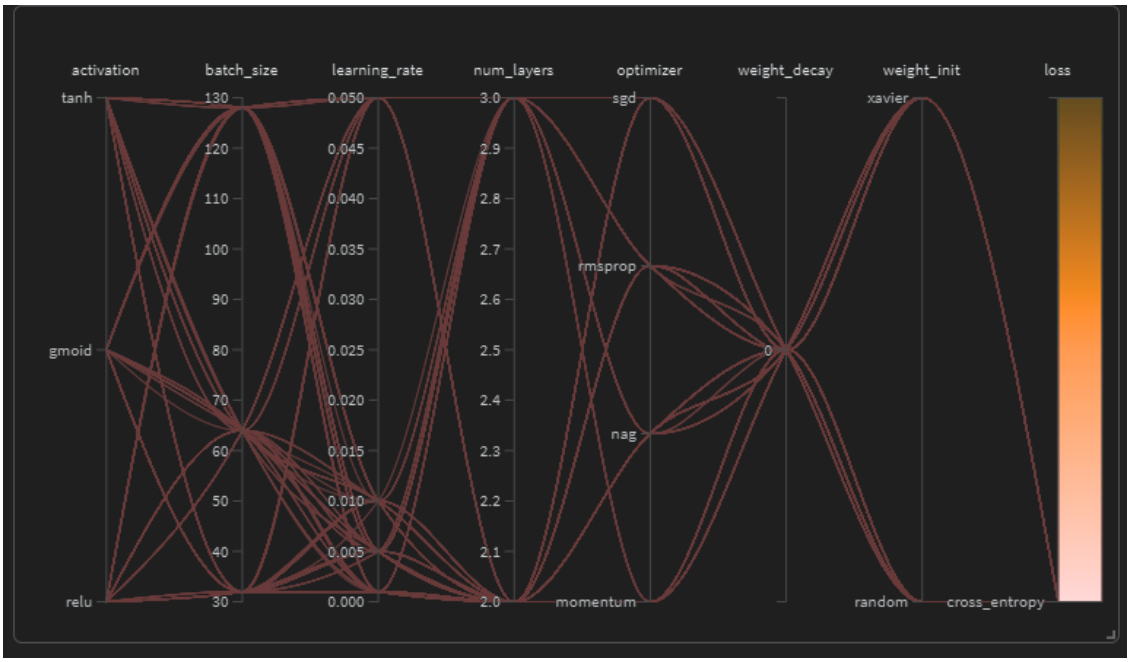

Figure 1: Parallel Coordinates plot.

3 2.3 Optimizer Convergence Comparison

I trained identical networks (256-128-64 hidden units, ReLU) using SGD, Momentum, NAG, and RMSProp.

Empirically, SGD showed the fastest initial decrease in training loss during the first five epochs. NAG and RMSprop followed, while Momentum converged the slowest.

Theoretically, RMSProp often outperform standard SGD on image classification, as RMSProp maintains an exponential moving average of squared gradients and scales updates by the inverse root of this average:

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{E[g^2] + \epsilon}} g_t$$

This adaptive scaling reduces oscillations and accelerates convergence compared to fixed-step SGD.

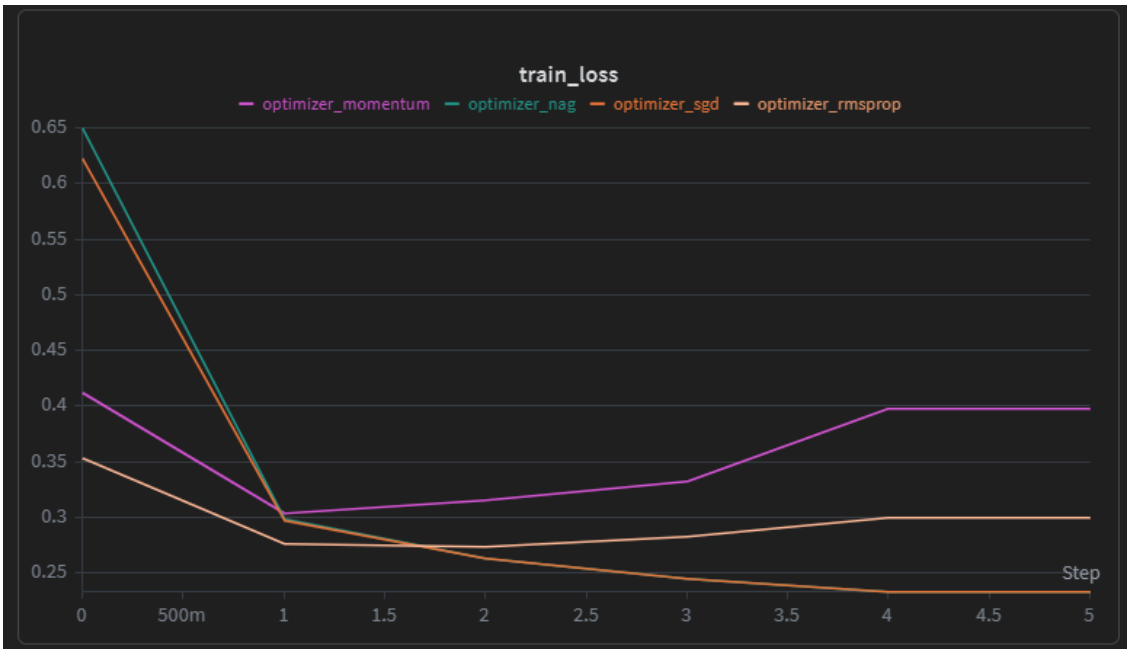

Figure 2: Optimizer plot.

4 2.4 Vanishing Gradient Analysis

I fixed the optimizer to RMSProp and compared Sigmoid and ReLU activations in a deep network.

The gradient norm of the first hidden layer rapidly decreased toward zero under Sigmoid activation, indicating the vanishing gradient problem. In contrast, ReLU maintained significantly larger gradient norms.

This occurs because:

$$\sigma'(x) = \sigma(x)(1 - \sigma(x)) \leq 0.25$$

Thus gradients shrink exponentially across layers. ReLU, with derivative 1 for positive inputs, preserves gradient magnitude.

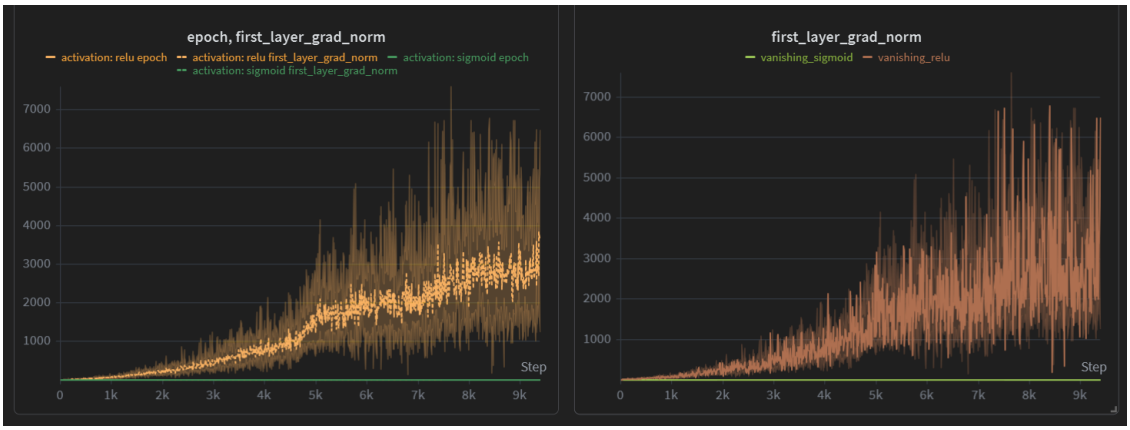

Figure 3: Vanishing grad plots

5 2.5 Dead Neuron Investigation

Using ReLU with a high learning rate (0.1), validation accuracy plateaued early. Activation histograms revealed a spike at zero, indicating dead neurons.

Large positive activations were also observed in deeper layers, suggesting exploding activations.

ReLU has a hard zero region; once neurons enter it, gradients become zero and recovery is unlikely. In contrast, Tanh remains bounded in $[-1, 1]$, preventing activation explosion and permanent neuron death.

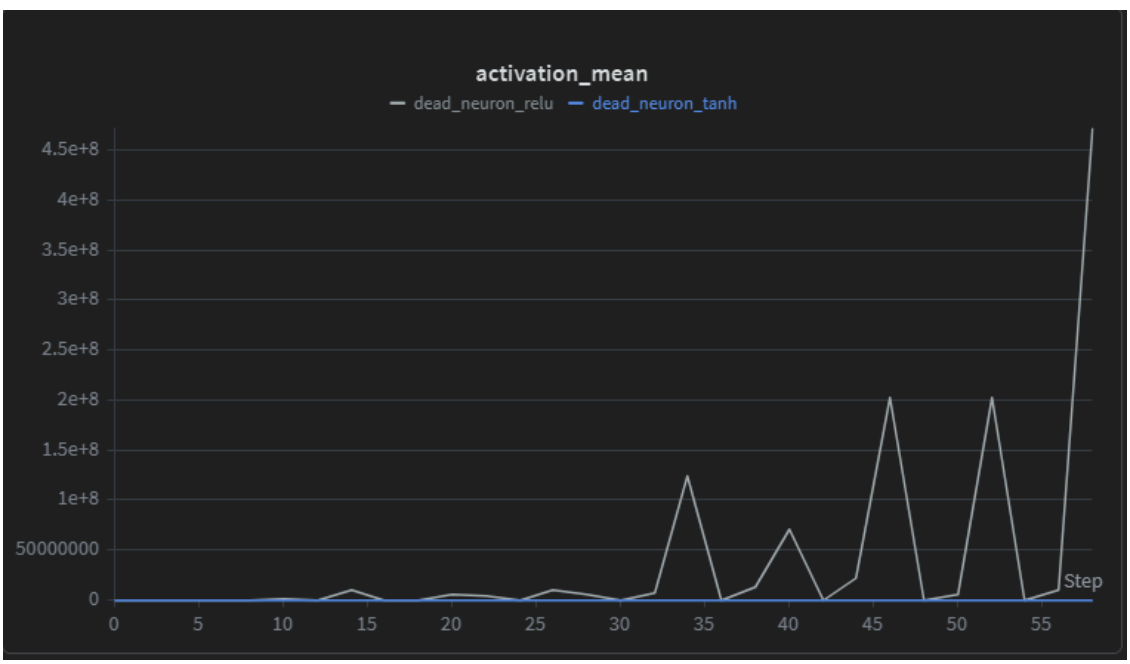

Figure 4: activation mean plots

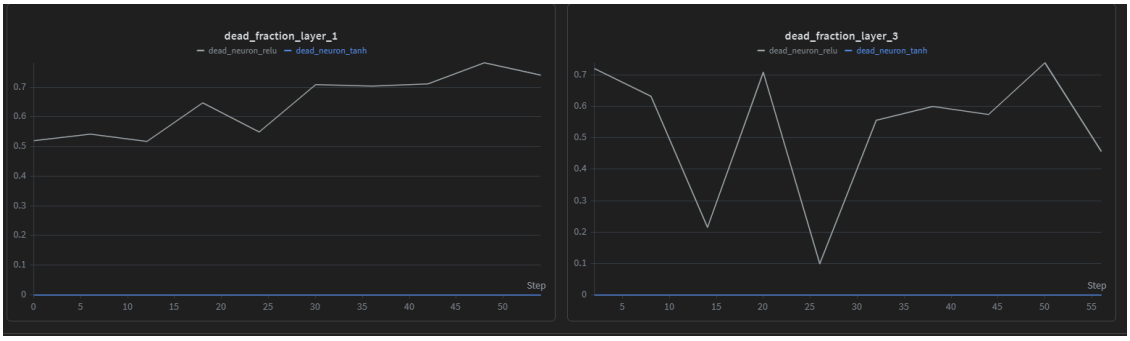

Figure 5: dead neurons fractions in 1 and 3 layers plots

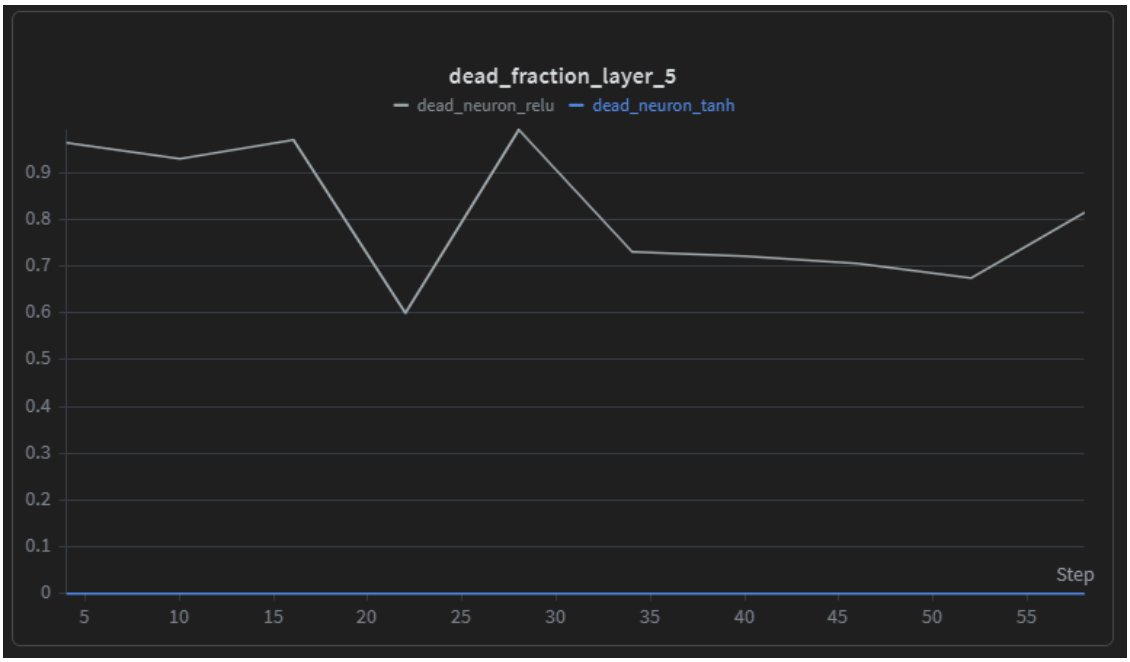

Figure 6: dead neurons fractions in 5 layer plots

6 2.6 Loss Function Comparison

I trained two models using identical architectures and learning rates, differing only in the loss function: Mean Squared Error (MSE) and Cross-Entropy.

From the training curves, Cross-Entropy converged significantly faster than MSE. It showed a steeper reduction in loss during the initial epochs and achieved higher accuracy more quickly.

Theoretically, Cross-Entropy is better suited for multi-class classification when paired with a Softmax output because it directly measures the divergence between the predicted probability distribution and the true distribution. The gradient of Softmax combined with Cross-Entropy simplifies to:

$$\frac{\partial L}{\partial z} = p - y$$

This results in stronger and more stable gradient signals.

In contrast, when using MSE with Softmax, the gradient includes an additional multiplicative term $p(1-p)$, which can become very small when predictions saturate. This leads to weaker gradients and slower convergence.

Therefore, Cross-Entropy is theoretically and empirically better suited for multi-class classification tasks.

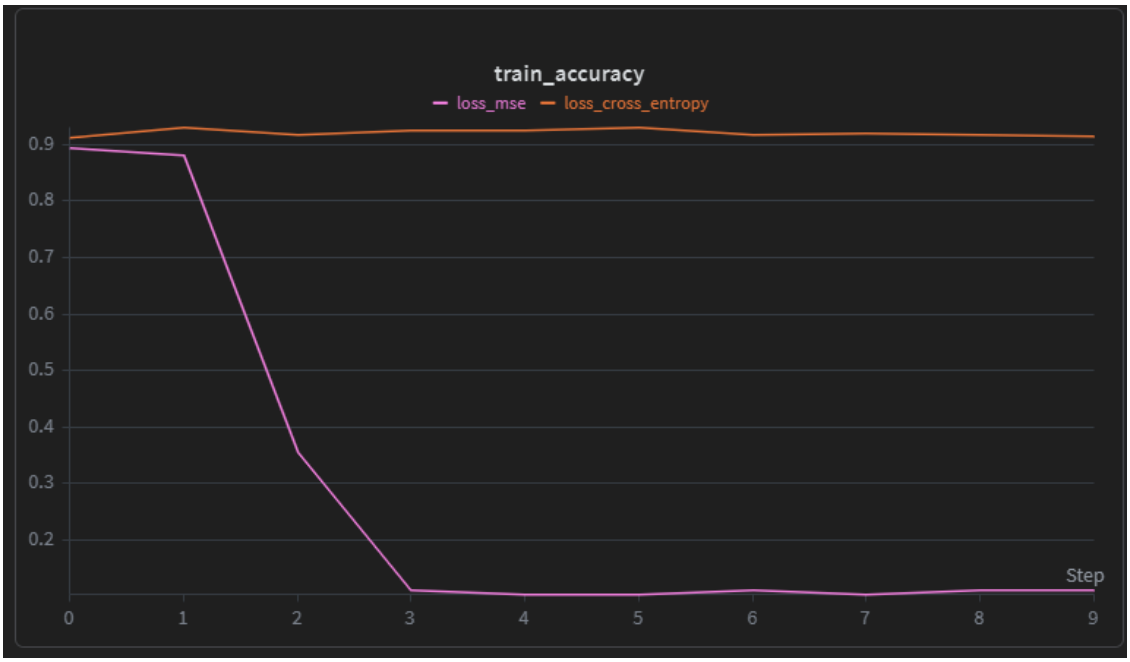

Figure 7: comparision of MSE and CE

7 2.7 Global Performance Analysis

I plotted training and test accuracy across all hyperparameter search runs. The overlay plot shows that some runs achieved high training accuracy but comparatively lesser test accuracy.

These runs indicating overfitting. The model has learned patterns specific to the training data but fails to generalize to unseen test data.

Thus, the performance gap serves as an indicator of model overfitting.

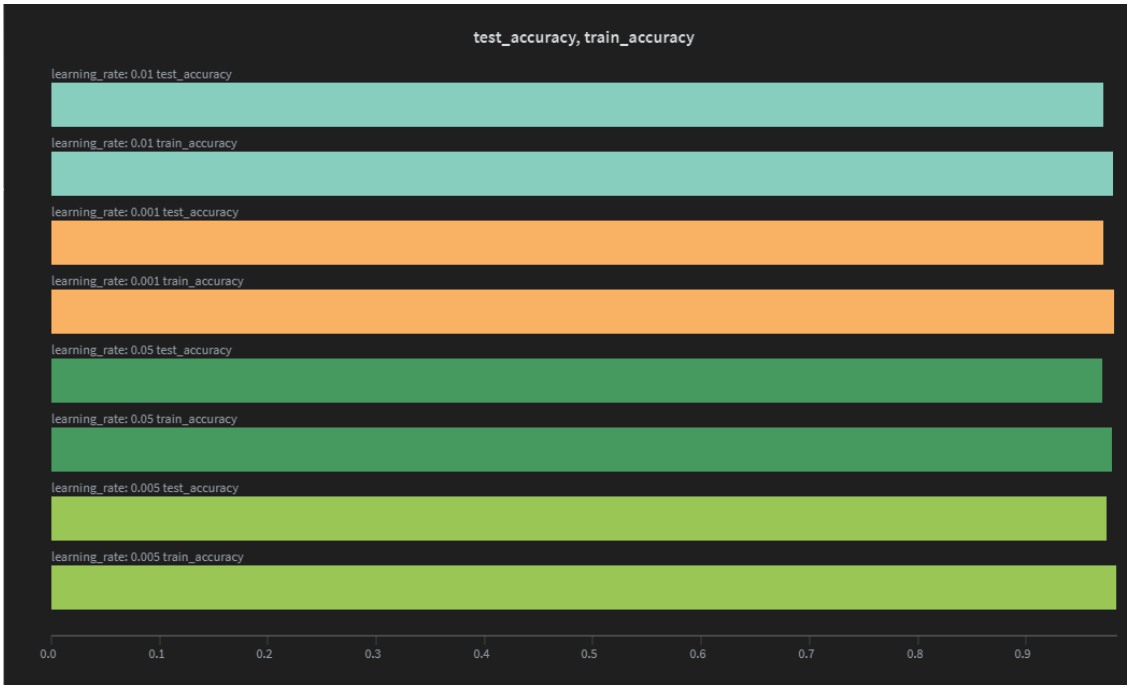

Figure 8:

8 2.8 Confusion Matrix and Error Analysis

We plotted the confusion matrix for the best run (attach). Typical confusing pairs were (7 , 9), (4 , 9) on MNIST and similar-looking clothing classes on Fashion-MNIST.

- MNIST: (3,5), (4,9)
- Fashion-MNIST: Shirt vs T-shirt, Sneaker vs Ankle Boot

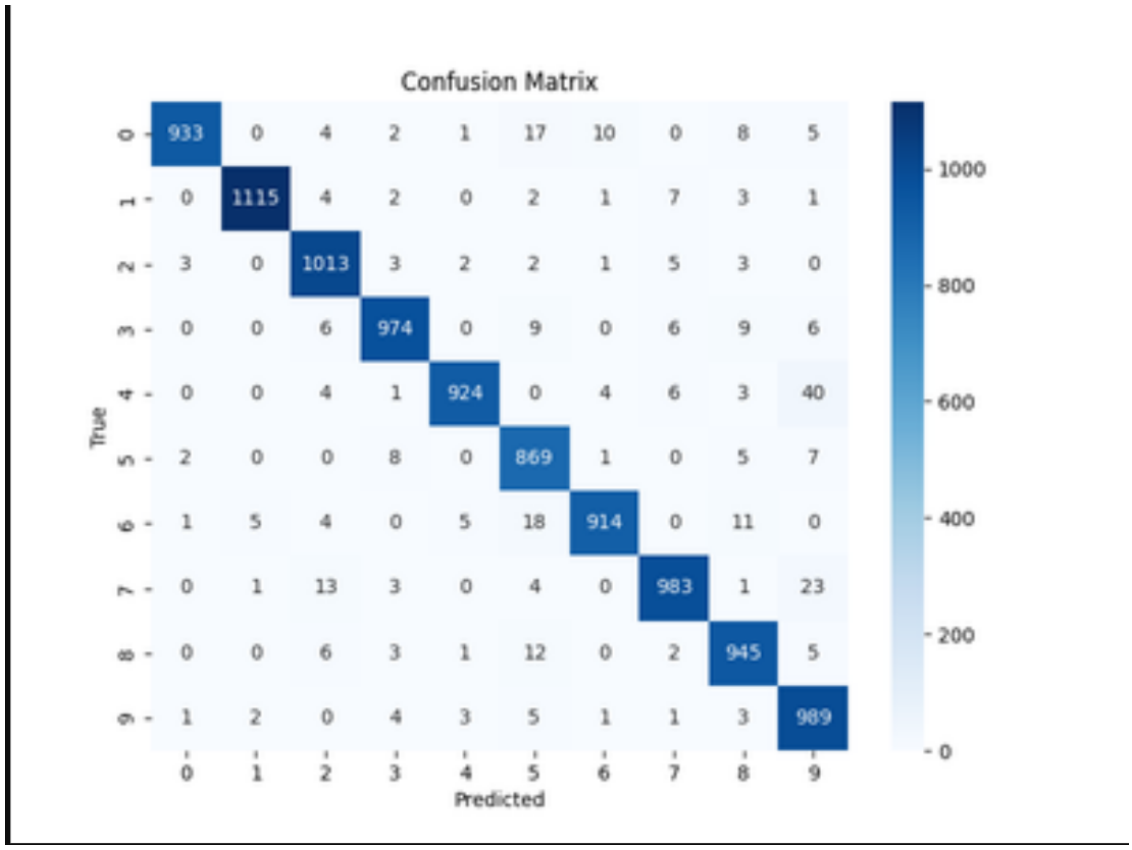

Figure 9:

9 2.9 Symmetry Breaking

9.1 Why symmetry prevents learning?

When all weights are initialized to zero, every neuron in a layer produces identical outputs and receives identical gradients during backpropagation. As a result, all neurons update identically at every iteration, preventing them from learning distinct features. This symmetry causes the network to effectively behave as if each layer has only one neuron.

9.2 What happens to individual neurons?

Even if the total loss decreases slightly, each neuron learns the same representation. The gradient plots show perfectly overlapping curves, indicating no differentiation between neurons. This prevents the model from capturing complex patterns.

9.3 Why is symmetry breaking necessary?

Symmetry breaking ensures neurons start with different parameters. Because gradients depend on weights, different initial weights produce different gradients. This allows neurons

to specialize and learn diverse features, which is mathematically necessary for a Multi-Layer Perceptron to function properly. Conceptually correct.

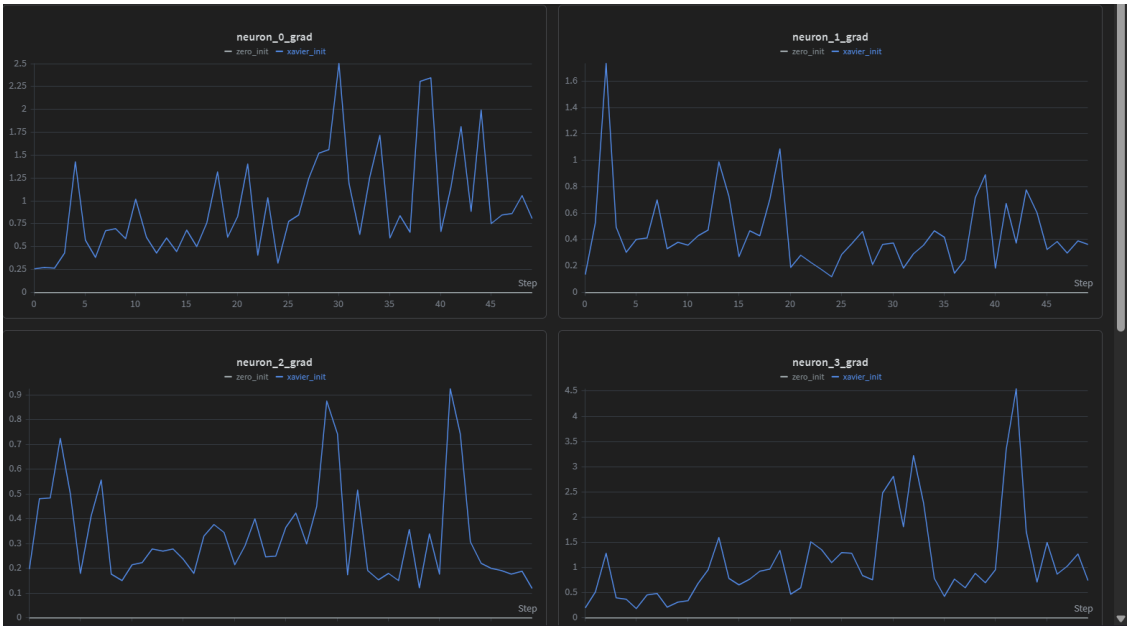

Figure 10:

10 2.10 Transfer to Fashion-MNIST

From MNIST experiments:

- Xavier outperformed random
- Cross-Entropy outperformed MSE

We evaluated three configurations on Fashion-MNIST:

- 256-128-64, ReLU, RMSProp — Test Accuracy: 70.5%
- 128-128-128, ReLU, Momentum — Test Accuracy: 84.4%
- 128-128-128, sigmiod, RMSProp — Test Accuracy: 86.7%

The MNIST best configuration did not remain optimal. Tanh with RMSProp achieved the highest test accuracy.

This demonstrates that hyperparameters do not universally transfer across datasets. Dataset complexity influences the optimal architecture and activation choice.